

# TRUTHLENS AI

## **AI-Powered Digital Image Investigation Platform using IBM watsonx.ai and Granite Foundation Models**

### **PROBLEM STATEMENT**

The rapid growth of digital media has increased the spread of manipulated images, misinformation, deepfakes, and forged digital evidence. Existing image verification methods often require expert knowledge, specialized forensic software, or manual investigation, making authenticity verification slow and inaccessible for many users.

Individuals, journalists, investigators, organizations, and cybersecurity professionals need a reliable system that can quickly analyze digital images, identify suspicious modifications, and generate understandable forensic reports. However, most available solutions either focus only on metadata analysis or only on visual inspection, resulting in incomplete investigations.

### **OBJECTIVE**

Develop an **AI-powered Digital Image Investigation Platform** that automatically analyzes uploaded images and generates intelligent forensic reports using **IBM watsonx.ai** and **IBM Granite Foundation Models**.

The system aims to:

- Detect possible image manipulation and authenticity issues.
- Extract and analyze image metadata.
- Perform visual inspection of uploaded images.
- Generate AI-powered investigation reports with risk assessment.
- Provide actionable recommendations for further verification.
- Deliver explainable results through an interactive web interface.

### **PROPOSED SOLUTION**

TruthLens AI adopts a **multi-agent architecture** where specialized agents collaboratively perform different stages of digital image investigation.

The platform integrates:

- Image Upload & Validation
- Metadata Analysis Agent
- Visual Analysis Agent
- IBM Granite AI Report Generation Agent
- Investigation Orchestrator
- Report Storage and Dashboard

IBM **watsonx.ai Runtime** hosts the **Granite-4-h-small** foundation model, which generates comprehensive forensic reports based on metadata and visual analysis findings.

### **TECHNOLOGIES USED**

- IBM Cloud
- IBM watsonx.ai Runtime
- IBM Granite-4-h-small
- Python
- FastAPI
- React
- TypeScript
- Pillow (PIL)
- Git & GitHub

### **Expected Outcome**

TruthLens AI provides an intelligent, automated, and explainable solution for digital image investigation. The platform assists users in identifying suspicious images, assessing authenticity risks, and generating professional investigation reports, making digital forensic analysis faster, more accessible, and reliable.

**Project Name:** TruthLens AI

**Domain:** Artificial Intelligence | Computer Vision | Digital Image Forensics

**Developed Using:** IBM watsonx.ai, IBM Granite Foundation Models, IBM Cloud